

Copilot POC

ADUSA CLOUD ENGINEERING

A close-up photograph of a person's hand, wearing a white sleeve, carefully plucking tea leaves from a lush green bush. The background is a soft-focus field of similar greenery.

Agenda

- High level Idea
- Tech Stack
- Repo Structure

High Level Idea

The goal is to build an **end-to-end intelligent automation** powered by **GitHub Copilot** that keeps the organization's **KeyVault Shared Module** (built on the AVM wrapper pattern) perpetually up-to-date, validated, compliant, and documented — with **zero manual intervention** from the engineering team.

The Core Problem Being Solved:

Every time Microsoft publishes a new version of the **Azure Verified Module (AVM) for Key Vault**, the internal shared module wrapper must be manually updated, tested, compliance-checked, documented, and reviewed — a repetitive, time-consuming, and error-prone process.

The Copilot-Powered Vision:

- Copilot **watches** for AVM Key Vault releases via the Public Bicep Registry (br/public:avm/res/key-vault/vault)
- It **understands the change diff** between the old and new AVM version
- It **intelligently updates** the wrapper module and example parameter file only where applicable
- It **validates** the update by deploying into a sandbox environment
- It **enforces governance** via SCF (Secure Cloud Foundation) policy compliance checks
- It **self-documents** any parameter-level changes in the README
- It **opens a PR**, summarizes the changes, and **notifies the team**



Tech Stack

Component	Tool/Technology
IaC Language	Azure Bicep
Module Standard	Azure Verified Modules (AVM)
Automation Engine	GitHub Actions
Copilot Integration	GitHub Copilot (Coding Agent)
Azure Deployment	Azure CLI / az deployment commands
Compliance Check	Azure Policy / az policy state
Notification	Microsoft Teams Webhook / GitHub Mentions
Documentation	Auto-generated Markdown README update
Version Detection	Bicep Public Registry API + br/public feed



Repo Structure

○ Code



```
keyvault-shared-module/
|
├── modules/
|   └── keyvault.bicep          ← AVM Wrapper Module
|
├── parameters/
|   └── keyvault.example.bicepparam ← Example Parameter File (Input)
|
├── tests/
|   └── deploy.validation.sh     ← Validation Deployment Script
|
├── .github/
|   └── workflows/
|       └── avm-update-automation.yml ← Main Copilot Automation Workflow
|
└── README.md                  ← Auto-updated documentation
```



Thank you

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